

AirSense

Campus Pilot Initiative

IoT-Powered Air Quality Monitoring, AI Forecasting and Campus Operational Decision Support

SUBMITTED TO

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PKR 20,000One-Time Budget
Requested**3 Months**Pilot Duration -
Autonomous**2
Campuses**BIC Islamabad + BIC
Karachi**100% In-
House**AI Models Built by BIC
Students

1. Why This Proposal Deserves Your Approval

This is not a request to fund a classroom exercise. This is a proposal to deploy a real, working, AI-powered IoT system on this campus, built entirely by BIC students, validated at an international level, and ready to go live today. AirSense is an environmental intelligence platform. It monitors air quality, forecasts it using machine learning models built in-house, and converts those forecasts into plain-language decisions that benefit the people on this campus. Every component of it was built here, by students from this institution.

We are asking for PKR 20,000. In return, BIC receives a live rooftop sensor feeding real data into AI models trained specifically on this campus, a publicly accessible air quality dashboard that carries the BIC name, three months of institutional-grade environmental data it owns outright, and the distinction of hosting Pakistan's first student-built, COIL AI-validated, IoT-driven campus air quality intelligence system. That is not a small thing. That is a landmark for what student-led innovation at BIC can produce.

One PKR 20,000 investment.

One sensor on one rooftop. Real data. Real AI. A public dashboard. A two-campus network.
This is what BIC students build when the institution believes in them.

2. Born from COIL AI: Internationally Validated, Ready to Deploy

2.1 Where AirSense Came From

AirSense was conceived, designed, and fully developed by BIC students during COIL AI, the international academic innovation programme that placed BIC in structured collaboration with partner institutions abroad. The programme challenged student teams to identify pressing real-world problems and build technology-driven solutions with genuine commercial and social impact. AirSense was BIC's answer: a system that does not just measure air quality, but forecasts it with AI and converts those forecasts into decisions.

The concept was not only developed during COIL AI. It was refined through iterative international peer review, expert critique, and structured evaluation. What emerged was a platform with a validated architecture, a working AI pipeline, and a clear deployment path. The campus pilot is not the beginning of AirSense. It is the moment AirSense stops being a validated concept and starts being a deployed product, on BIC's own rooftop.

AirSense: Developed During COIL AI

COIL AI was an internationally structured innovation programme connecting Beaconhouse International College with partner universities abroad. Student teams were challenged to build applied, real-world technology solutions, evaluated by international academic panels. AirSense was developed by BIC's team within this programme and refined through structured international review.

2.2 The Problem We Are Solving and Why It Matters Now

PM2.5 is the most harmful category of airborne particulate matter. Invisible to the naked eye and small enough to pass through the lungs directly into the bloodstream, it is directly linked to respiratory illness, cardiovascular disease, reduced cognitive performance, and long-term organ damage at sustained exposure levels. Islamabad and Karachi both experience significant seasonal, traffic-driven, and construction-driven PM2.5 elevations. Yet localised, real-time, campus-level data is effectively nonexistent in both cities.

The consequence of this gap is that schools, colleges, and institutions make daily decisions about outdoor activities, physical education sessions, ventilation management, and campus events without any data about what their students and staff are actually breathing. AirSense closes this gap. For BIC specifically, deploying the system means management has, for the first time, a factual environmental basis for these decisions. That is not an incremental improvement. It is a transformation in institutional environmental awareness.

3. The AirSense Platform: Where IoT Meets Artificial Intelligence

3.1 Real IoT. Real AI. Real Deployment.

The word IoT is used extensively in academic programmes across Pakistan. In most cases it describes a concept taught in a classroom. In this proposal, IoT describes something specific, physical, and operational: a sensor unit mounted on the BIC Islamabad rooftop, measuring the air every five minutes, transmitting data over the campus network, feeding it into machine learning models built by BIC students, and returning a forecast that helps the campus make better decisions. This is not IoT as a subject. This is IoT as a functioning deployed product.

Platform Layer	Component	What It Delivers for BIC
IoT Sensing	Physical Rooftop Sensor	Live PM1, PM2.5, PM10, temperature, humidity, pressure, and rainfall readings every five minutes from the BIC rooftop. This is real, locally measured data about your specific campus, not regional estimates or external proxies.
Data Backbone	ESP32 + Wi-Fi + SD Card	Every reading stored locally on an SD card and transmitted hourly to the campus workstation via existing campus Wi-Fi. Dual-channel redundancy: data is never lost even if connectivity drops temporarily.
In-House AI	Four BIC-Built ML Models	XGBoost, Random Forest, Gradient Boosting, and Isolation Forest models, all developed during COIL AI and ready for deployment. Retrained on live BIC campus data during the pilot. Owned entirely by the project and the institution.

Decision Layer	AI Decision Advisor	Converts forecast numbers into plain-language operational guidance. Not raw data. Decisions: hold the outdoor session, adjust the HVAC, reschedule the event, or proceed without concern.
Public Dashboard	Live Web Interface	A publicly accessible, BIC-branded dashboard showing live air quality, 72-hour forecasts, health risk levels, and operational recommendations. Accessible from any browser. Visible to the world.

Every layer of this platform was designed, built, and tested by BIC students. No external AI vendor. No licensed software. No third-party model. This is BIC's intellectual product, and the campus pilot is where it becomes BIC's deployed product.

4. In-House AI Models: Built by BIC Students, Owned by BIC

4.1 The Models Are Ready. They Just Need BIC Data.

One of the most compelling facts about this proposal is what the team is not asking for. The AI models that power AirSense were not purchased, licensed, subscribed to, or sourced from any external platform. They were built from scratch during COIL AI by the AirSense team. They are functional, tested, and ready for deployment. The campus pilot does not create these models. It trains them on real BIC campus data, making them campus-specific and substantially more accurate for this environment than any generic model could be.

#	Model	Function	Business Value for BIC
1	XGBoost	PM2.5 Prediction Engine	Forecasts PM2.5 at 1, 3, 6, and 24-hour horizons. The core engine behind every alert, recommendation, and planning decision the system produces. After campus retraining, it knows this specific rooftop, this specific environment.
2	Random Forest	Health Risk Classifier	Classifies current and forecast conditions into risk tiers for students, staff, and outdoor workers. Drives the health advisory panel that tells management exactly which population groups are at risk and by how much.
3	Gradient Boosting	Multi-Horizon Forecaster	Extends the forecast window to 24 hours with high accuracy. Enables proactive scheduling: management knows tomorrow morning's air quality before tonight ends, and can plan accordingly.
4	Isolation Forest	Anomaly Detection	Identifies unusual PM2.5 spikes that deviate from predicted patterns. Flags events such as nearby construction, generator activity, or fire incidents that require immediate campus attention rather than a scheduled response.

4.2 The Retraining Advantage

All four models were trained on publicly available datasets during the COIL AI development phase. That training was sufficient for the models to be validated and for the platform architecture to be confirmed as sound. However, a model trained on generic data is less accurate than a model trained on the specific environment it is deployed in. When the pilot sensor goes live on the BIC rooftop, the team retrains every model on actual BIC Islamabad data. Within weeks, the system knows the difference between a Tuesday morning rush hour and a Friday afternoon. It knows how the campus microclimate responds to changing seasons. It forecasts for BIC, not for a generic city average.

At the end of the pilot, BIC owns four campus-calibrated AI models. They were not purchased. They were not licensed. They were built by BIC students and made more accurate by BIC data. That is a genuine institutional asset.

5. A Two-Campus Initiative: BIC Islamabad and BIC Karachi Building Together

5.1 One System. Two Cities. One Institution.

AirSense is not a single-campus project that has been extended a courtesy mention to Karachi. It is a genuinely collaborative initiative between Beaconhouse International College Islamabad and Beaconhouse International College (BIC) Karachi, with dedicated project leads at each campus, a shared codebase, shared hardware specifications, and a unified development and deployment plan that treats both campuses as equal contributors to the same system and the same dataset.

From BIC Islamabad, Munim Qureshi leads system architecture, AI model development, and dashboard engineering, supervised by Ms. Sahifa Alam. From BIC Karachi, Areesha Aqeel leads the Karachi sensor deployment, local data collection, and campus-specific calibration, supervised by Mr. Sajid. Both project leads work from the same codebase, benchmark their results against each other, and iterate together on model improvements. This is what genuine cross-campus technical collaboration looks like, and it is happening at BIC.

BIC Islamabad	BIC Karachi
Lead: Munim Qureshi Supervisor: Ms. Sahifa Alam • Primary pilot and production site. Hosts the live dashboard and data pipeline. • Leads system architecture, AI model development, and platform engineering. • Inland climate, traffic patterns, academic calendar signatures in data. • Budget management and procurement handled from Islamabad.	Lead: Areesha Aqeel Supervisor: Mr. Sajid • Parallel deployment at BIC Karachi using identical hardware, firmware, and pipeline. • Generates a simultaneous Karachi dataset enabling direct two-city model comparison. • Coastal climate, sea-proximity effects, different industrial and traffic profiles. • Validates the portability and scalability of the AirSense system across Pakistani cities.

5.2 Why Two Campuses Makes This Stronger for BIC

Running simultaneously in Islamabad and Karachi, AirSense collects data from two fundamentally different urban environments across the same period. This allows the team to confirm whether models trained in one city generalise meaningfully to another, directly informing commercial deployment planning. It also means the public dashboard, from launch, shows two Pakistani cities, not one. That is a more compelling demonstration, a stronger academic output, and a more powerful statement about what BIC as an institution produces.

Crucially, the two-campus structure means this is a BIC network achievement, not only a BIC Islamabad project. Both campuses gain the benefit of the data, the models, and the dashboard. Both campus supervisors, Ms. Sahifa Alam and Mr. Sajid, contribute to the academic oversight of the project. The pilot completion report, submitted at the end of three months, is a joint deliverable representing both institutions. This kind of cross-campus collaboration is genuinely rare and reflects well on the entire BIC organisation.

6. Deployment Plan: From Approval to Live System in Ten Days

The team is ready to move the moment approval is received. Hardware has been specified, vendors confirmed, firmware written, and the installation plan finalised. The complete path from budget approval to a live rooftop sensor with data flowing into AI models is ten days, structured across five clear phases with no dependencies outside what BIC already has in place.

Phase	When	What Happens
Procurement	Days 1-5	- Orders placed immediately with Embedded Studio, Digilog.pk, Electrobes, A.E Solution, Clopal Online, and local Islamabad suppliers. - All vendors identified and verified in advance. Delivery estimated within three to five business days. No new sourcing required. - Radiation shield sourced locally in Islamabad. Quotation confirmed within 48 hours of approval.
Assembly	Days 6-8	- Full bench assembly: PMS7003 sensor, BME280 environmental sensor, ESP32 microcontroller, SD logging module, rain sensor, and power supply integrated into weatherproof IP65 enclosure. - ESP32 firmware flashed and all sensor outputs verified numerically. SD card logging confirmed. Wi-Fi transmission to campus workstation tested successfully. - Complete bench validation before any rooftop access. No installation proceeds until the full unit is confirmed operational indoors.
Installation	Days 9-10	- Sensor unit mounted at the agreed rooftop location using stainless hose clamps. Power cable routed and all enclosure penetrations sealed with RTV silicone. - Campus Wi-Fi extender repositioned for rooftop signal coverage. Signal strength confirmed before leaving the rooftop. - End-to-end validation: live readings visible on dashboard, SD card backup active. Pilot clock starts. Data begins accumulating immediately.

Collection	Months 1-3	- Fully autonomous operation. No daily human intervention required. The system runs through weekends, holidays, and the summer academic break without interruption. - Weekly data quality checks by the team. Any anomalies investigated, logged, and resolved promptly. Public weather feed synchronised hourly for wind speed and direction data. - Break-period data is scientifically valuable: it provides a clean, low-activity baseline against which term-time patterns are measured and modelled.
Training and Reporting	Month 2 onwards	- Feature engineering begins in Month 2 as sufficient data accumulates. All four AI models retrained on live BIC campus data using walk-forward validation. - Decision framework tested in pilot mode throughout Month 3. Recommendations logged and benchmarked against observed conditions. - Full pilot completion report submitted to management and supervisors at both campuses at the end of Month 3, with dataset, models, and Phase 2 commercial roadmap.

7. The AirSense Dashboard: BIC's Environmental Intelligence, Made Public

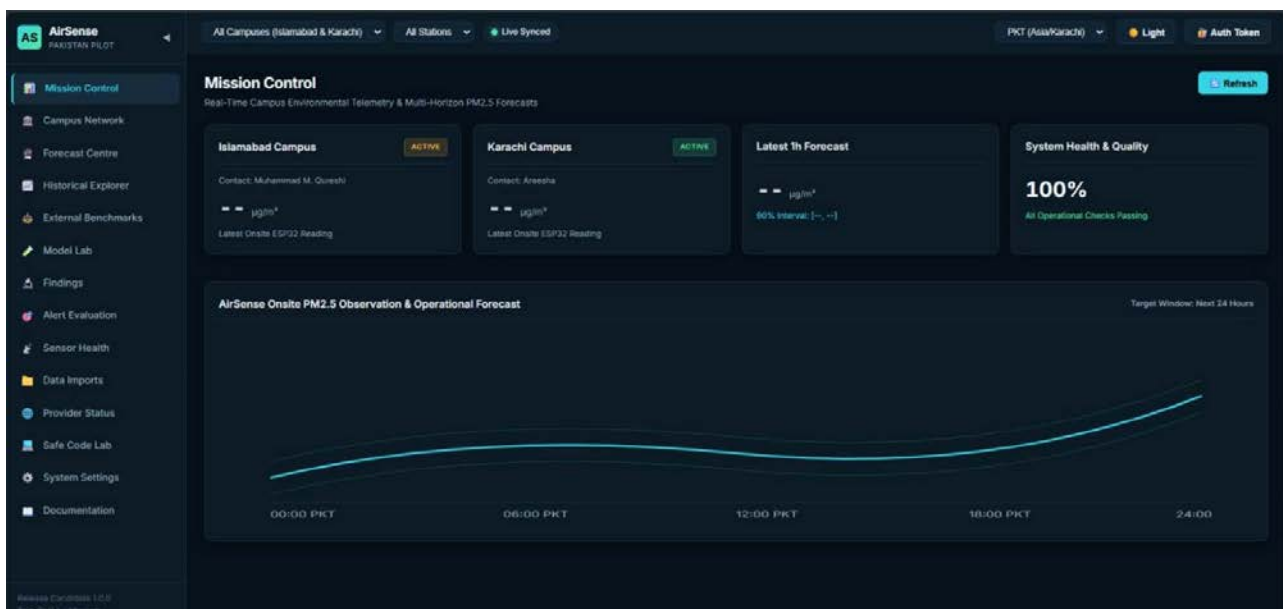
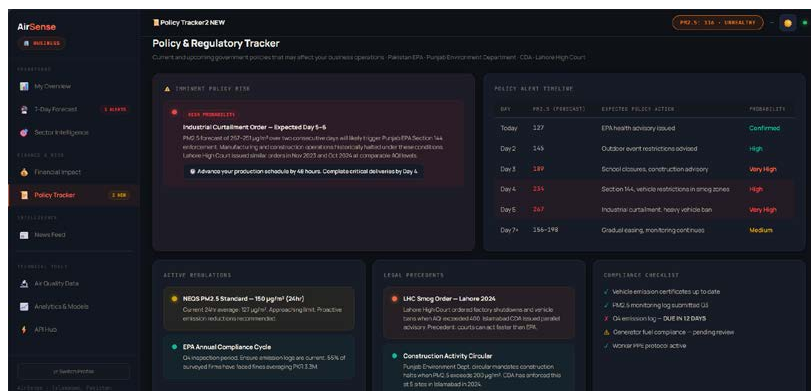
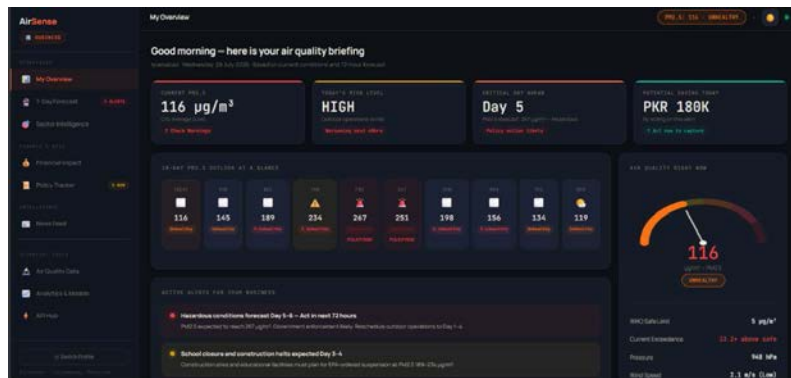
The AirSense dashboard is where all the data, forecasts, and decisions become visible. During the campus pilot it runs on a campus workstation. After the pilot, it is deployed publicly on the web, accessible from any browser, carrying the BIC name, showing live data from both campuses, and making BIC's environmental intelligence visible to the world. This is not a backend system that only the team can see. It is a public-facing product that anyone can open.

The AirSense dashboard will be publicly accessible on the web.

Anyone can open it and see, in real time, what the air quality is in Islamabad and in Karachi. It carries the BIC name. It is BIC's contribution to public environmental awareness in Pakistan.

7.1 Main Dashboard: Live AQI and Station Status

The landing screen displays the current AQI reading, live PM2.5 concentration, a colour-coded air quality status, and the campus sensor status panel. Designed to be readable at a glance by any stakeholder from a student to a member of senior management, with no technical knowledge required.



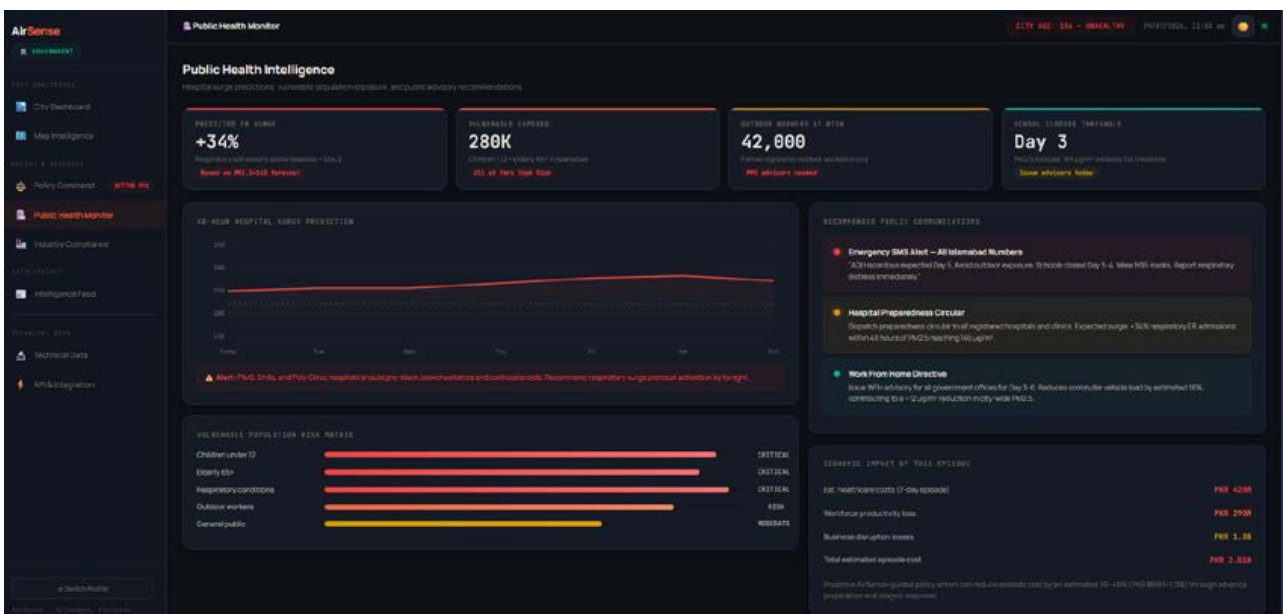
7.2 AI Decision Advisor

Management or teaching staff can ask directly in plain language: Is it safe to hold the sports day outdoors this Saturday? Should we adjust ventilation this afternoon? The advisor ingests the live forecast automatically and responds with a grounded, actionable recommendation. No data science knowledge required to use it.



7.3 Health Risk Panel and Operational Window Calculator

The health risk panel classifies conditions by user group: students, staff, and outdoor workers, with specific risk levels and guidance for each. The operational window calculator identifies the lowest-pollution periods in the next 24 hours, enabling proactive scheduling before a problem arrives rather than reactive response after it does.



8. Operational Decision Making: Why This Is the System That Matters

8.1 Data Is Not the Product. Decisions Are.

There are many tools that measure air quality. There are very few that tell you what to do about it. This distinction is the entire commercial and institutional value of AirSense. Knowing that PM2.5 is currently 75 micrograms per cubic metre is a data point. Knowing that it will reach 120 in the next three hours, that outdoor physical activity should be suspended for afternoon classes, that the safest window for the sports day tomorrow is between 7:00 and 9:00 in the morning, and that HVAC systems on the upper floors should switch to recirculation mode within the next hour, that is a decision. AirSense produces the second kind of output. Every time. Automatically.

Who Benefits	What Triggers the System	What AirSense Recommends
Campus Management	PM2.5 forecast rising sharply in next 3 hours	Move the outdoor assembly indoors. Notify PE teachers before the afternoon session. Adjust ventilation on upper floors. Send a health advisory to parents if sustained.
Facilities Management	High PM2.5 combined with low wind dispersion forecast	Switch HVAC to recirculation mode. Reduce fresh-air intake. Schedule outdoor maintenance for the lower-pollution window identified for the following morning.
Health and Safety Officers	Sustained exceedance of WHO or Pakistan EPA thresholds	Generate an automated alert log. Notify responsible staff. Adjust outdoor activity schedules for students with respiratory conditions. Log the exceedance for compliance records.
Event Planners	Outdoor event scheduled within 24-hour forecast window	Identify the lowest-pollution two-hour window within the day. Recommend timing. Flag if no safe outdoor window exists and suggest indoor relocation with lead time.

8.2 Scope: What AirSense Can Offer Beyond the Campus

The campus pilot is where AirSense proves itself. The market beyond the campus is where it scales. Pakistan's logistics operators, construction companies, healthcare networks, corporate real estate managers, and educational institutions across the country represent a substantial addressable base for a system that converts air quality forecasts into sector-specific operational decisions. No equivalent locally-built, locally-deployed, AI-powered product currently exists in this space.

Market Segment	The Gap Today	What AirSense Offers
Educational Institutions	Schools and colleges scheduling outdoor activities without environmental data	Campus-specific air quality dashboards, activity scheduling support, health advisory integration. BIC is the reference deployment and the proof of concept.
Logistics and Delivery	Delivery fleets operating across polluted urban corridors with no dispatch optimisation	Dispatch window optimisation, driver health protection, route air quality mapping, ESG documentation support for corporate sustainability reporting.
Construction and Infrastructure	Construction sites managing dust and worker safety with no predictive data	Dust episode forecasting, outdoor work-window optimisation, regulatory compliance logging for Pakistan EPA dust management requirements.

Healthcare Facilities	Hospitals and clinics without advance warning of air quality episodes affecting patient intake	24-hour ahead episode alerts, patient outdoor activity management, clinical staff pre-positioning for respiratory intake surges on high-PM2.5 days.
Corporate and ESG	Companies with growing environmental reporting obligations and no local data source	Automated air quality logging, threshold exceedance records, and exportable data for ESG and Health, Safety and Environment (HSE) compliance documentation.

The commercial market for environmental decision intelligence in Pakistan is largely unaddressed. AirSense, validated through COIL AI and proven at BIC campuses, is positioned to be the first locally built product to serve it. And it was built here, at this institution.

9. What This Means for BIC: The Full Institutional Return

This section answers the question that matters most to management: what does BIC actually get from approving this proposal? The answer is significant across multiple dimensions. The list below is specific, tangible, and lasting. Each item is a direct consequence of this project being approved and deployed.

What BIC Gains	Why It Matters
A Public Dashboard Carrying the BIC Name	The AirSense dashboard will be publicly accessible, branded with BIC, and visible to anyone with a browser. Parents, prospective students, partner institutions, and the broader public can see that BIC deployed a student-built AI system. This is institutional reputation built in real time, for PKR 20,000.
Real-Time Environmental Data for This Campus	For the first time, BIC management, health officers, and faculty have access to real-time, campus-specific air quality data. Decisions about outdoor activities, physical education sessions, and ventilation are no longer made without information. They are made with it.
Four Campus-Calibrated AI Models	At pilot completion, BIC owns four retrained AI models calibrated to the Islamabad campus environment. These models were built by BIC students and made accurate with BIC data. They are not dependent on any subscription or external service. They are the institution's own AI assets.
Three Months of Owned Environmental Data	A quality-controlled, timestamped dataset of BIC Islamabad's air quality over three months. This is an academic and research asset with value beyond this project. It can support student dissertations, departmental research, and institutional reporting. BIC owns it outright.
AI Programme Credibility and Proof	A deployed, publicly visible AI system built by students is the strongest possible evidence that BIC's CSSE and AI programmes produce graduates capable of building real products. It validates the curriculum in a way that no course description or academic transcript can replicate.
A Cross-Campus Innovation First	BIC Islamabad and BIC Karachi will have jointly built and deployed a real-world AI system. This is a cross-campus technical collaboration that is genuinely unprecedented within the BIC network and reflects directly on the strength of the institution's inter-campus academic culture.
Commercial Incubation Opportunity	If AirSense succeeds commercially, BIC is its incubator. The institution that believed in the project before it had revenue, customers, or a public track record.

	That association is the kind of thing that defines innovative institutions and attracts students, faculty, and partners who want to be part of what comes next.
Student Careers Advanced	Munim Qureshi and Areesha Aqeel graduate with documented, verified experience deploying a real IoT system, training AI models on live data, and managing a two-city cross-campus technical project. That portfolio distinguishes a candidate in any engineering or technology hiring process.

10. Budget: PKR 20,000 - Every Rupee Accounted For

10.1 The Full Itemised Procurement Plan

This is not an estimate. Every item below has a vendor, a model number, a confirmed price, and a role in the system. The total requested is PKR 20,000, which covers all 16 hardware components, the locally sourced radiation shield, and a contingency buffer for minor price movements at point of order. No cloud costs. No subscriptions. No recurring fees during the pilot. One payment. Three months of live operation.

#	Component	Vendor	Qty	Cost (PKR)
1	PMS7003 PM1/PM2.5/PM10 Laser Particle Sensor	Embedded Studio	1	4,200
2	ESP32 WROOM-32D Development Board	Digilog.pk	1	1,160
3	BME280 Temperature / Humidity / Pressure Sensor	Electrobes	1	850
4	Arduino MicroSD Card Reader Module	Electrobes	1	150
5	Samsung EVO Plus 16 GB MicroSD Card	Daraz.pk	1	1,450
6	Waterproof IP65-rated Electrical Enclosure	A.E Solution	1	750
7	Jumper Wire Kit, 120 pcs	Daraz.pk	1	440
8	5V 2A AC/DC Power Adapter	Electronics Hub	1	170
9	Black Nylon Self-Locking Cable Ties, 100-pack	Daraz.pk	1	120
10	2-inch Stainless-Steel Hose Clamps, 2 pcs	DreamsMart.pk	1	60
11	Raindrop Detection Sensor Module A	Digilog.pk	1	160
12	Rain Drop Moisture Detection Sensor Module B	Electrobes	1	150
13	UNI-T UT363 Digital Anemometer (wind speed validation)	Electrobes	1	4,250
14	Clopal 10 m Heavy-Duty Power Extension Lead	Clopal Online	1	3,295

15	GMSA RTV Silicone Weatherproof Sealant, 310 ml	Expert Tools World	1	390
16	Mounting Nuts and Bolts M3-M6, 8 pcs	Multan Electronics	8	192
Hardware Subtotal (Items 1-16)				PKR 17,787
Radiation Shield (local sourcing) + Contingency Reserve				PKR 2,213
TOTAL BUDGET REQUESTED One-Time Capital Expenditure, No Recurring Costs				PKR 20,000

The buffer between the hardware subtotal (PKR 17,787) and the total requested (PKR 20,000) covers the radiation shield quotation from a local Islamabad supplier and any minor price movements at time of order. Any unspent portion will be returned to the institute at the conclusion of procurement.

10.2 Future Phases: What Is Not Being Asked For Today

Phase	Description	Cost	Request Status
Phase 1	Hardware, installation, and pilot at BIC Islamabad. This proposal.	PKR 20,000 One-time	REQUESTED NOW
Phase 2	Model refinement, two-campus dataset comparison, public dashboard deployment.	Negligible Uses Phase 1 assets	No additional cost
Phase 3	Live cloud pipeline, API integrations, commercial SaaS launch in Pakistan. Separate proposal following successful pilot.	PKR 5,000-10,000/month	Separate proposal post-pilot

11. What BIC Receives at Pilot Completion

No.	Deliverable	Description
1	Validated Campus Air Quality Dataset	Three months of timestamped, quality-controlled hourly records: PM1, PM2.5, PM10, temperature, humidity, pressure, and rain flags from the BIC Islamabad rooftop. BIC's first owned campus environmental dataset.
2	Four Retrained Campus AI Models	XGBoost, Random Forest, Gradient Boosting, and Isolation Forest models retrained on live BIC campus data. Documented performance metrics per forecast horizon. No external dependencies. Owned by BIC.
3	Live Public Dashboard	A publicly accessible, BIC-branded web dashboard displaying live air quality, 72-hour forecasts, health risk levels, and AI-generated operational recommendations from both BIC campuses. Operational at pilot conclusion.
4	Decision Framework Evaluation Report	Pilot-mode assessment of the operational decision layer: accuracy of triggered recommendations, correct alert rate, false-alert analysis, and a refined action playbook based on real BIC campus conditions.

5	Full Pilot Completion Report	A formal report submitted to Ms. Saba Ahson and Ms. Sahifa Alam covering system performance, data quality, model accuracy, two-campus comparison findings, public dashboard status, and a Phase 3 commercial roadmap.
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12. Conclusion: A PKR 20,000 Investment in Something That Will Succeed

AirSense was built during COIL AI. It was validated internationally. The models are ready. The hardware is specified. The deployment plan is finalised. Two project leads, supervised by two department heads across two BIC campuses, are ready to execute. The only thing this project is waiting for is an institution that believes in it enough to give it the budget and the rooftop it needs.

We are that team. This is that project. And we are asking BIC to be that institution.

A PKR 20,000 approval does not just fund a sensor. It funds a deployed AI system, a public dashboard, a validated dataset, four retrained machine learning models, a two-campus collaboration report, and the foundation for a commercial product that was designed and built by students at this institution. Every single one of those outcomes flows directly from this one approval. The return on this investment is substantial, concrete, and lasting.

1	Approval of a one-time hardware budget of PKR 20,000 for procurement of all components as itemised in Section 10.
2	Permission to mount the weatherproofed sensor assembly at an agreed location on the BIC Islamabad rooftop for a period of three months. Fully reversible. No structural modifications required.
3	Facilitation of access to a rooftop power point and repositioning of a campus Wi-Fi extender for sensor connectivity. No new infrastructure required.
4	Recognition of this as an officially supported BIC student initiative, enabling appropriate access to campus computing resources for data processing and model training during the pilot period.

The sensor goes on the roof. The data comes in. The AI models learn.

The dashboard goes live. The world sees what BIC students built.

We built AirSense at COIL AI. We are asking BIC to be the place where it becomes real.

We believe this project will succeed. We are asking the institution to believe it too.

Respectfully submitted,

Munim Qureshi

Project Lead, BIC Islamabad | AirSense

August 2026

Approval Sign-Off

Role and Name	Campus	Signature	Date
Head of Institute Ms. Saba Ahson	BIC Islamabad		
Head of CSSE/AI Ms. Sahifa Alam	BIC Islamabad		
Project Lead Munim Qureshi	BIC Islamabad		